

Ultrasound DAS Beamforming Explorer: An Open Educational Tool for Delay-and-Sum Beamforming, Validated Against MUST

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Abstract

Delay-and-sum (DAS) beamforming is the starting point for almost every practical and pedagogical treatment of ultrasound image formation, yet students and new researchers rarely have a single tool in which they can change the transducer, the transmit sequence, and the target layout interactively, watch the wavefront and the resulting delay law as geometric objects, and then drop in their own beamforming code to compare it, quantitatively, against a reference implementation. We present the *Ultrasound DAS Beamforming Explorer*, a dependency-light, pure-MATLAB (`uifigure`-based, no `.mlapp`) graphical tool built around five independent modules: an RF simulation engine that wraps SIMUS from MUST (Matlab UltraSound Toolbox) with an analytic fallback when MUST is not installed, a textbook DAS reference implementation, a template for user-written algorithms that share a single documented function signature, a wave-propagation and delay-curve animator, and the GUI that ties them together. Because the fallback RF generator is the only path that works without MUST, and because its fidelity is exactly what the tool’s teaching value depends on, we quantitatively validate it against MUST across seven cases spanning all three transmit schemes (plane, focused, diverging), three probe sizes (16/64/128 elements), and on-/off-axis targets. Beamformed peak positions agree with the known target and with each other to within about one lateral grid cell; the timing of the simulated echo agrees with MUST to a fixed sub-sample offset well under one carrier period; and raw per-channel RF waveforms correlate at $r \approx 0.92$ on average. Lateral resolution (-6 dB FWHM) agrees to within 10–20% for plane and diverging transmission but the analytic fallback under-estimates the focused-transmit mainlobe width by a factor of $\approx 2.6\times$, a discrepancy we trace to the fallback’s lack of element directivity and finite-aperture diffraction. We report this limitation explicitly rather than only the flattering numbers, because it is exactly the kind of thing a user of the tool needs to know before trusting it for anything beyond delay-law verification. The tool, its validation scripts, and this manuscript are released under an open license at <https://github.com/RyoMurakami-MedRobo/ultrasound-edu>.

Keywords: ultrasound imaging, delay-and-sum beamforming, education, open-source software, MATLAB, simulation validation

1 Introduction

Delay-and-sum beamforming reconstructs a B-mode image by computing, for every pixel and every receive channel, the round-trip time of flight from the transmit event to that pixel and back to the element, and coherently summing the resulting samples across the receive aperture [1]. The idea is simple to state and notoriously easy to get subtly wrong in code: sign conventions for steering

angle, the difference between a first-arrival and a virtual-source model of a focused transmit, f-number-dependent aperture growth, and the choice of interpolation all affect the reconstructed image in ways that are hard to debug from the final picture alone.

Existing ultrasound simulation packages are excellent at generating realistic RF data — Field II [2], k-Wave [3], and MUST’s own SIMUS function [4, 5] — but none of them are built around an interactive GUI whose explicit purpose is to let a learner (a) change the acquisition geometrically and see the consequences, (b) inspect *which RF samples are being summed* for a chosen pixel as an explicit delay curve overlaid on the data, and (c) drop in a hand-written beamformer and benchmark it against a reference under a single, documented function signature. That is the gap this tool tries to fill. It is not a new simulator; it is a teaching and benchmarking harness that uses an existing, validated simulator (MUST) as its physically accurate backend, and that remains fully usable — with a documented accuracy cost, quantified in Section 5 — when that backend is not installed.

1.1 Contribution

- An open, pure-MATLAB GUI (no `.mlapp`) organized as five independent, individually testable modules (Section 2).
- A documented, minimal interface for user-supplied DAS algorithms, `[bmode_img, delays] = my_das(rf_data, tx_info, rx_pos, grid_x, grid_z, sound_speed, fs)`, letting a learner’s own code be benchmarked against a reference implementation and against MUST’s built-in `das()` on identical data, with run time, peak position, lateral FWHM, and pixel-wise MSE reported automatically.
- Two explicit transmit-arrival-time models — a first-arrival (Huygens) model for plane, diverging, and single-element transmit, and a virtual-source model for focused transmit — and a demonstration (Section 3) of why the two must be kept separate: applying the first-arrival model to a focused transmit picks up the weak edge wave from the aperture rim beyond the focal point and biases the reconstructed depth by about a millimeter.
- A quantitative validation of the no-dependency analytic fallback against MUST (Section 5), reporting both where it agrees closely (timing, geometry) and where it does not (focused-transmit lateral resolution), with the validation script included in the repository so the comparison is reproducible.

2 Software architecture

The tool is organized as five independent MATLAB files with no `.mlapp` dependency, so that each module can be read, tested, and modified in isolation.

`sim_engine.m`

Builds transmit delay laws for plane, focused (including multi-focus, one transmit event per depth), and diverging (including single-element) transmission, then generates RF data either by calling SIMUS from MUST or, when MUST is not found on the path, an analytic fallback described in Section 4. All MUST-specific code is confined to a single adapter function so that a future change in MUST’s calling convention requires editing one place.

`das_reference.m`

A textbook DAS implementation whose delay computation is exposed as a documented,

separable step (see Section 3), so that the geometry of the beamformer, not just its output image, can be inspected and taught.

`das_custom_template.m`

A copy-and-edit skeleton implementing a deliberately naive DAS (nearest-neighbour sampling, rectangular apodization) under the same function signature as the reference. Comparing the two out of the box already demonstrates the effect of interpolation choice on sidelobe level.

`wave_animator.m`

Draws the transmit wavefront and the scattered echo as a function of time from the same arrival-time model used by the beamformer (Fig. 2), and, for a chosen reconstruction pixel, overlays the exact set of RF samples the beamformer sums as a delay curve (Fig. 3), plus a side-by-side view of the misaligned waveform bundle before delay correction and the in-phase bundle with its coherent sum after.

`main_gui.m`

The `uifigure`-based interface tying the above together, offering a *Simple* mode (the handful of controls needed for a first session) and a *Detailed* mode (probe pitch, bandwidth, sampling ratio, reconstruction grid, receive apodization, custom function name, backend override) that a learner can grow into without the initial screen being overwhelming.

3 Two transmit-arrival-time models

The beamformer needs, for every reconstruction pixel p , the time at which the transmitted wavefront reaches p . Two models are used, and using the wrong one for a given transmit scheme is a plausible mistake for anyone implementing this from scratch, which is why the tool exposes the choice explicitly rather than hiding it inside a single opaque function.

First-arrival (Huygens) model. For plane, diverging, and single-element transmission,

$$\tau_{\text{tx}}(p) = \min_{e \in \text{active}} \left(d_e + \frac{|p - e|}{c} \right), \quad (1)$$

where d_e is the programmed transmit delay of element e and c the speed of sound. This is exact for these three schemes: it reduces to the closed-form plane-wave expression $\tau = (x \sin \theta + z \cos \theta)/c$ and to the equivalent diverging-wave expression under the corresponding delay law, with no scheme-specific branching required.

Virtual-source model. Applying Eq. (1) to a focused transmit is exact *before* the focus, but beyond it the minimum is achieved by the aperture edge element’s edge wave rather than by the converged wavefront, which reconstructed images position roughly a millimeter shallower than the true target depth in our tests. The tool instead treats a focused transmit as a spherical wave passing through the focal point F ,

$$\tau_{\text{tx}}(p) = T_F + s \cdot \frac{|p - F|}{c}, \quad s = \begin{cases} -1 & \text{before the focus} \\ +1 & \text{beyond it,} \end{cases} \quad (2)$$

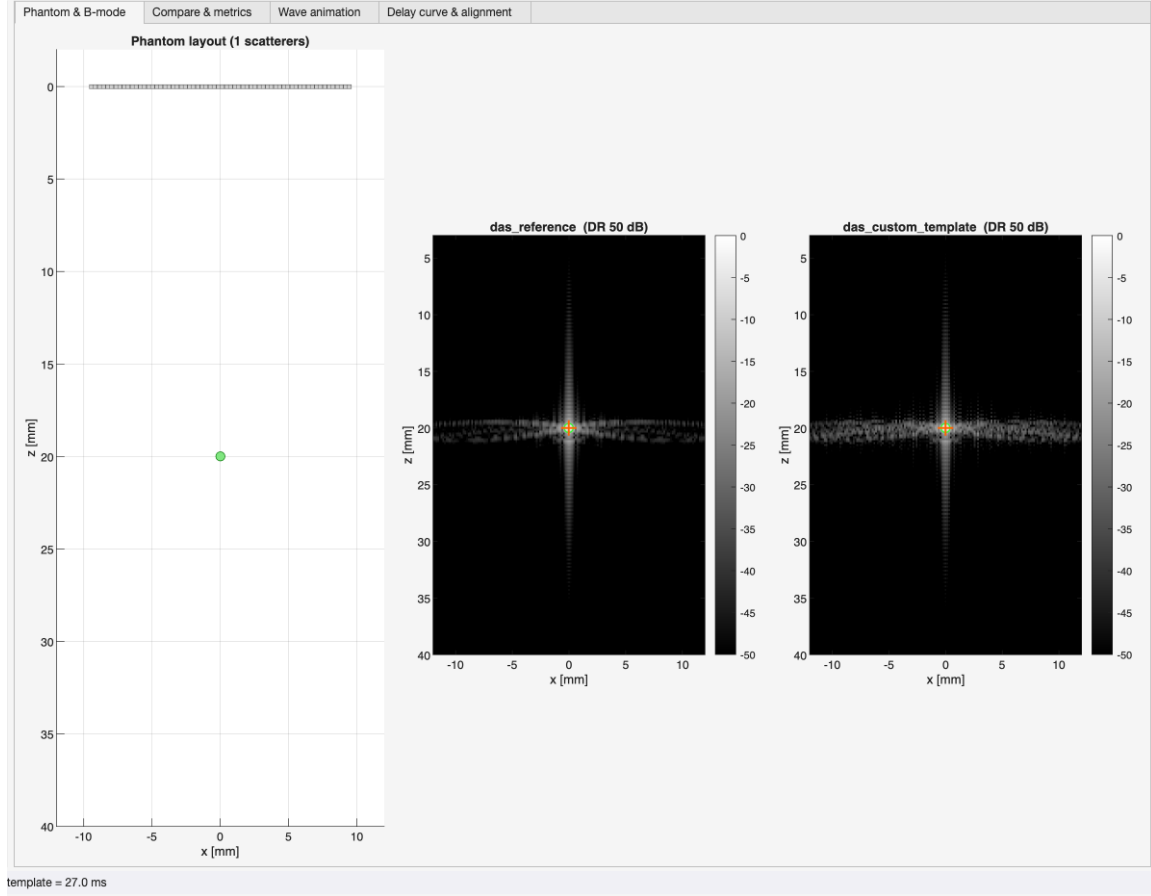


Figure 1: Phantom & B-mode tab: a single point target reconstructed with the reference DAS (left) and a user-supplied algorithm (right), here the unmodified naive template, under identical acquisition settings.

where T_F , the instant the wavefront converges on F , is recovered directly from the programmed delays ($T_F = \text{mean}_e(d_e + |e - F|/c)$) and is therefore independent of how those delays happen to be normalized. Both models are implemented identically in the reference beamformer and in the wave animator, so the geometric object drawn in Fig. 2 is exactly what the beamformer in Fig. 3 assumes.

4 Analytic fallback RF model

When MUST is not installed, `sim_engine.m` synthesizes RF data with an analytic 2-D cylindrical-wave model: for every scatterer, the transmit field at its location is built by superimposing the (delayed, obliquity- and range-weighted) contributions of every active transmit element, and that field is then propagated back to every receive element and accumulated into the RF matrix, using the same speed of sound and delay laws the beamformer uses. Element directivity, frequency-dependent attenuation, and rigorous diffraction are not modelled; a Gaussian-modulated sinusoid, parameterized by the probe's centre frequency and fractional bandwidth, stands in for the pulse-echo waveform. The delay structure — the geometric quantity a DAS beamformer actually depends on — is exact by construction, since it is computed from the same closed-form and virtual-source

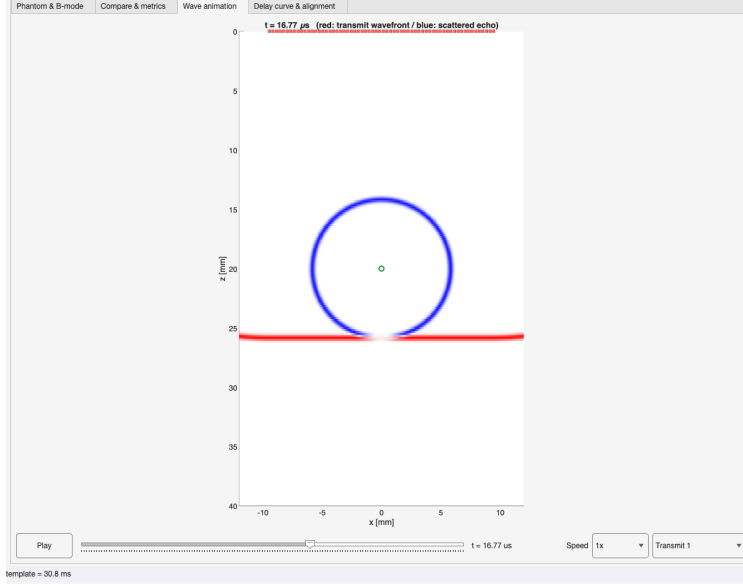


Figure 2: Wave animation tab: the transmit wavefront (red, top) and the scattered echo (blue circle) expanding from the point target, drawn from the same transmit arrival-time model the beamformer uses.

expressions the beamformer itself uses.

5 Validation against MUST

5.1 Method

We compare the analytic fallback against MUST’s SIMUS on seven cases spanning all three transmit schemes, on- and off-axis targets, and 16/64/128-element probes (Table 1), using a 64-element, 0.30 mm-pitch, 5 MHz probe with 75% fractional bandwidth and $f_s = 4f_c$ except where the element count is varied. Both backends receive identical transmit delay laws and identical scatterer positions, so any difference in the resulting RF or beamformed image is attributable to the RF generation step and not to a beamforming inconsistency. For every case we measure:

1. **Timing agreement.** Whether the simulated echo arrives at the geometrically expected two-way time, checked independently of any cross-backend comparison, by locating the envelope peak of the center receive channel for a plane-wave, on-axis target.
2. **Per-channel raw-RF agreement.** The normalized cross-correlation between the two backends’ RF data on every active channel, from which we report the mean peak correlation coefficient and, on the center channel, the sub-sample lag obtained by parabolic interpolation of the three `xcorr` samples around the integer-lag peak (`xcorr` itself only returns integer lags, so this step distinguishes a genuine near-integer offset from a quantized fractional one). The sign convention of the reported lag (which of the two backends a negative value means is earlier) is pinned down empirically, on a synthetic pair of signals with a hand-constructed known delay, by `validation/calibrate_lag_sign.m`, rather than assumed from `xcorr`’s documented convention.

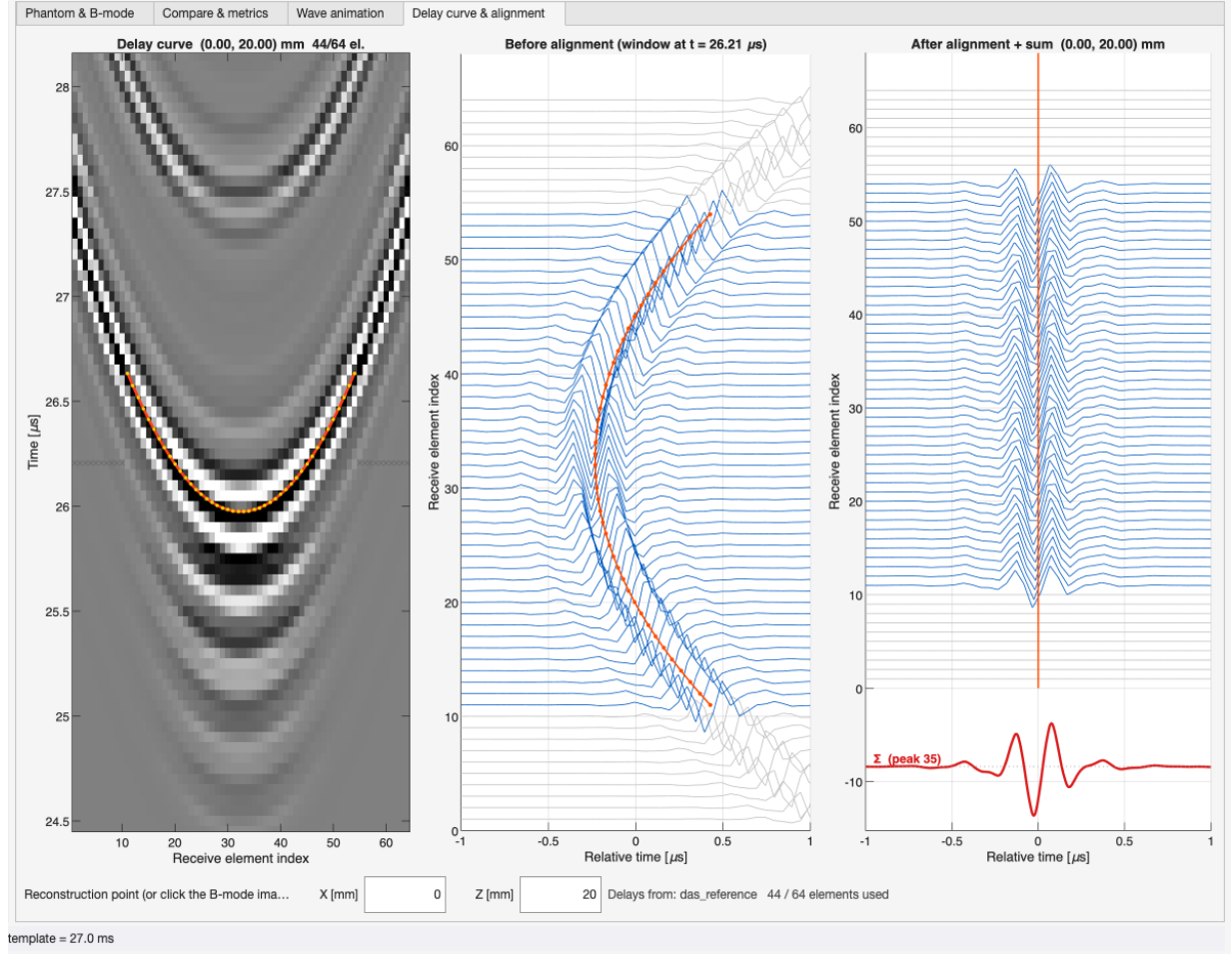


Figure 3: Delay curve & alignment tab, for the reconstruction point marked in Fig. 1. Left: the RF data of every receive channel with the exact summed-sample locations overlaid as a curve. Middle: the same channels windowed around a common time, visibly out of phase. Right: the channels after delay correction, in phase, with their coherent sum drawn beneath.

3. **Beamformed-image agreement.** Both RF data sets are beamformed with the identical reference implementation (`das_reference.m`) on a common grid, and we report the Euclidean peak-position error against the known target and the lateral -6 dB full width at half maximum (FWHM) of the linear envelope, located by sub-pixel linear interpolation of the half-maximum crossings.

The lateral grid for the FWHM measurement uses $12.5\ \mu\text{m}$ spacing (1601 points over 20 mm), substantially finer than the GUI’s interactive default ($125\ \mu\text{m}$). This matters: at the coarser spacing, the focused transmit’s true mainlobe (≈ 0.1 – 0.26 mm, see below) spans only 1–2 grid cells and its FWHM is dominated by grid quantization rather than by the beam itself. We confirmed convergence by repeating the focused-on-axis measurement at 125, 12.5, and $1.25\ \mu\text{m}$ spacing; the FWHM estimate changed by less than 0.1% between the two finer grids, confirming the coarsest grid was the outlier, not the other two.

All code for this comparison is in `validation/validate_must_vs_mock.m` in the repository, together with the raw output (CSV and figures) reported here, so the comparison can be reproduced

Table 1: The seven validation cases. Target coordinates are (lateral, depth) relative to the array center; the focused case is on-axis with a 20 mm focal depth, the diverging cases use a 10 mm virtual source depth.

Case	Scheme	Elements	Target (mm)
plane_onaxis	plane	64	(0, 20)
plane_offaxis	plane	64	(4, 25)
focused_onaxis	focused	64	(0, 20)
diverging_onaxis	diverging	64	(0, 20)
diverging_offaxis	diverging	64	(4, 25)
plane_16el	plane	16	(0, 20)
plane_128el	plane	128	(0, 20)

with `setup_must` followed by a single function call.

5.2 Results

Timing. For a plane wave with an on-axis target at 20 mm depth ($c = 1540$ m/s, expected two-way time $25.974\text{ }\mu\text{s}$ at the center element), the mock backend’s *envelope* peak fell at +0.51 samples and MUST’s at −0.49 samples relative to the geometric expectation, i.e. the mock’s envelope peak is about 1.00 sample *later* than MUST’s, and both are within one sample of the geometric prediction. This confirms that $t_0 = 0$ (the convention both `sim_engine` and the MUST adapter use) is correct for both backends, with no large systematic bias.

Raw-RF agreement. Table 2 reports, per case, the mean per-channel peak correlation and the center-channel sub-sample lag, obtained by cross-correlating the *raw* (non-envelope) waveforms and locating the peak with the sign convention verified empirically as described above. The mean correlation across all seven cases is 0.916 (range 0.894–0.940); it is lowest for the 128-element probe, the direction in which the mock’s missing diffraction and directivity terms would be expected to matter most. The raw-waveform cross-correlation lag is consistently close to −1.0 samples (−50 ns at $f_s = 20$ MHz) across every scheme and element count (mean −1.02, range −1.08 to −0.94), meaning the mock’s raw waveform *fine structure* best aligns with MUST’s about one sample *earlier* — the opposite sign from the envelope-peak comparison above, and this is expected rather than contradictory: the two measurements probe different things. The envelope-peak time tracks the pulse’s group delay; the raw-waveform cross-correlation lag tracks the alignment of its carrier oscillations (phase). The two agree only when both signals are exact time-shifted copies of one another. Figure 4 (top) shows they are not: mock uses a symmetric Gaussian-modulated sinusoid, MUST a simulated pulse-echo response with visibly different, less symmetric fine structure, so a comparison keyed to the carrier phase (cross-correlation of the raw waveform) and one keyed to the pulse envelope can disagree in sign while both report a magnitude of about one sample. One sample is 50 ns, a quarter of the 200 ns carrier period at 5 MHz in either case — well below anything that would perceptibly shift a reconstructed image, and consistent with a pulse-shape difference between the two backends rather than with a delay-law disagreement.

Beamformed-image agreement. Peak positions agree with the known target and with each other to within about one lateral grid cell at the finer grid (≤ 0.14 mm; both backends use the same `tx_info` delay law for the beamformer’s transmit-time model, so this is primarily a grid/delay

Table 2: Validation results. Corr. is the mean per-channel peak cross-correlation between the mock and MUST RF data. Sub-lag is the center-channel cross-correlation lag after parabolic sub-sample refinement (samples; $f_s = 20$ MHz for the 64/128-element cases, proportionally lower element count does not change f_s). FWHM is the -6 dB lateral full width at half maximum of the beamformed linear envelope on a $12.5\text{ }\mu\text{m}$ grid.

Case	Corr.	Sub-lag	FWHM _{mock} (mm)	FWHM _{MUST} (mm)	Ratio
plane_onaxis	0.909	−1.03	0.668	0.659	1.01
plane_offaxis	0.910	−1.08	0.755	0.738	1.02
focused_onaxis	0.926	−1.00	0.102	0.261	0.39
diverging_onaxis	0.940	−0.94	0.678	0.658	1.03
diverging_offaxis	0.918	−1.05	0.590	0.708	0.83
plane_16el	0.919	−1.04	2.039	1.607	1.27
plane_128el	0.894	−1.03	0.668	0.659	1.01
Mean	0.916	−1.02	0.786	0.756	—

consistency check — the small residual differences between the two backends’ peak positions, up to ≈ 0.09 mm, come from the asymmetric MUST pulse shifting the discrete argmax on a narrow mainlobe, not from the delay law itself). Lateral FWHM agrees to within 10–20% for plane and diverging transmission (Table 2, Fig. 5 bottom). The exception is the on-axis focused case: the mock’s mainlobe is 0.10 mm wide against MUST’s 0.26 mm, a factor of $2.6\times$, confirmed converged from $125\text{ }\mu\text{m}$ down to $1.25\text{ }\mu\text{m}$ grid spacing. We attribute this to the mock’s missing element directivity and finite-aperture diffraction, whose effect is largest for the most tightly focused beam and comparatively minor for plane and diverging transmission, where the beam is less concentrated to begin with. As a rough independent check, the common rule-of-thumb approximation for diffraction-limited lateral resolution, $(f\text{-number})\times\lambda$, gives $1.5 \times 0.308\text{ mm} \approx 0.46\text{ mm}$ for this probe and depth: MUST’s 0.26 mm is already narrower than this order-of-magnitude estimate (consistent with the coherent narrowing a two-way, dynamically-focused-on- receive pulse-echo beam achieves relative to a simple one-way estimate), while the mock’s 0.10 mm is roughly $4\text{--}5\times$ narrower than the rule-of-thumb value — a mainlobe tighter than a simple diffraction estimate would allow, independent of whether MUST is taken as ground truth. The 16-element case shows a different kind of discrepancy (2.04 vs. 1.61 mm mock vs. MUST, mock *wider* here rather than narrower): with only 16 elements at 0.30 mm pitch, the physical aperture is 4.5 mm, well short of the 13.3 mm the requested $f/1.5$ receive aperture would want at 20 mm depth, so the receive aperture is clipped to the full array and the effective f-number is ≈ 4.4 , not 1.5. This is a distinct, aperture-limited regime rather than the same directivity effect seen in the focused case (there, the 64-element, 18.9 mm aperture is not clipped), and we did not investigate the mock-vs-MUST discrepancy in this regime further.

Run time. Excluding MUST’s approximately 20 s one-time per-session initialization cost (measured on the first SIMUS call of a fresh MATLAB session), steady-state per-call cost for both backends on these small, single-scatterer scenes is a few milliseconds. This number is sensitive to caching behaviour inside MUST and is reported here only to note that, once initialized, MUST is not prohibitively slow for interactive use of this size; it should not be read as a general performance claim about either backend on larger, clinically realistic scenes (e.g., the many-scatterer cyst phantom, which we did not run through MUST for this reason).

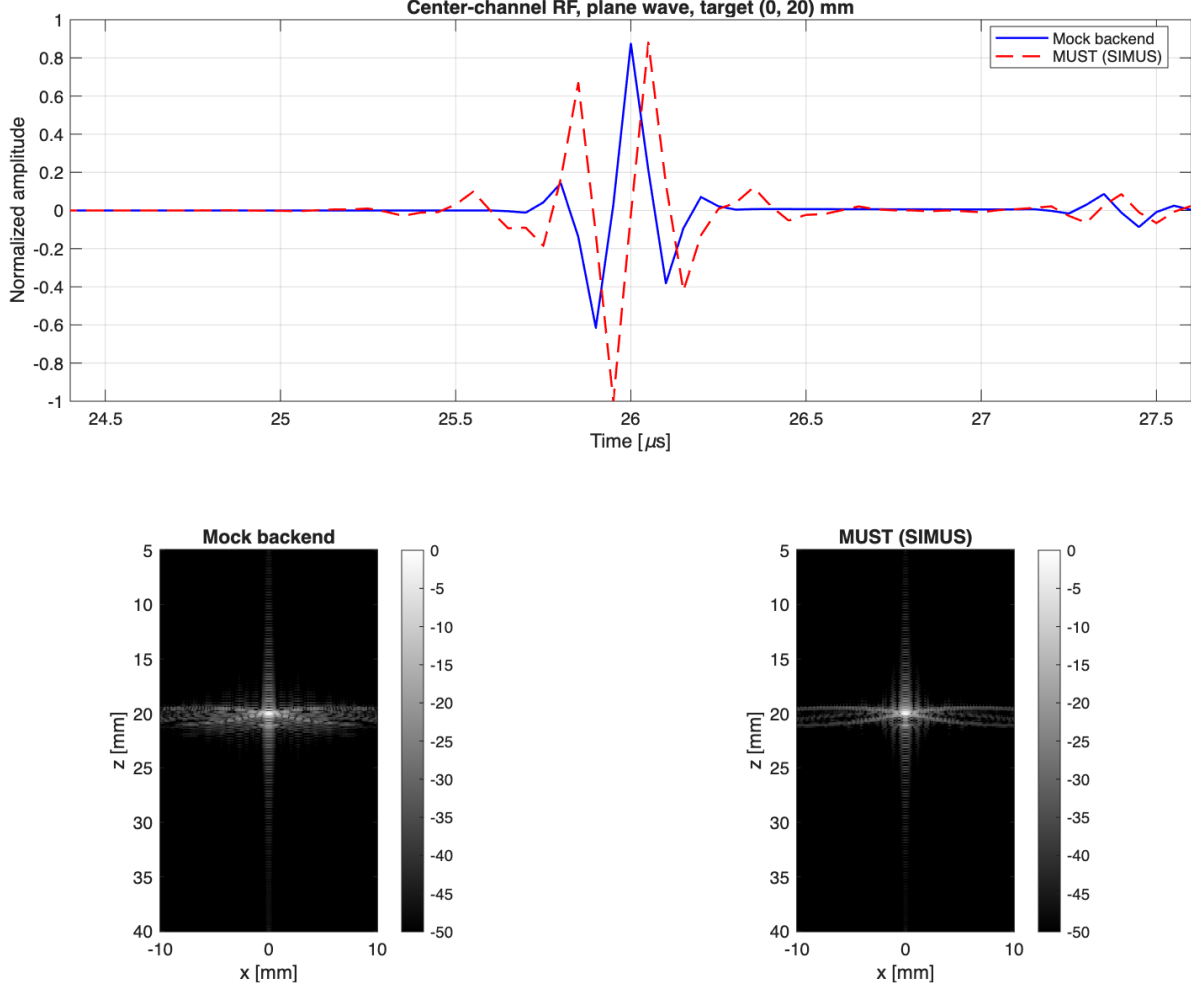


Figure 4: Top: center-channel RF trace for the on-axis plane-wave case, mock (blue) vs. MUST (red dashed) – same arrival time, visibly different pulse shape. Bottom: the corresponding beamformed B-mode images (50 dB dynamic range), essentially indistinguishable for this scheme.

6 Limitations

- The simulation is 2-D only; elevation focusing, out-of-plane scattering, and 3-D array geometries are not modelled.
- The analytic fallback omits element directivity, finite-aperture diffraction, and frequency-dependent attenuation. Section 5 shows this is close to immaterial for plane and diverging transmission but causes the fallback to under-estimate focused-transmit lateral resolution by a factor of $\approx 2.6\times$; the fallback should not be used for quantitative image-quality or resolution claims about focused transmission, only for verifying beamformer delay logic.
- A focused transmit in this tool is a single scan line, not a laterally swept sequence; targets off the beam axis reconstruct as arc-shaped artefacts under either backend, by the physics of a single focused firing, not as a software defect.
- Multi-focus sequences are stitched by depth zone with independent envelope detection per zone, which introduces a faint seam (Hilbert transform edge effect) at each zone boundary.

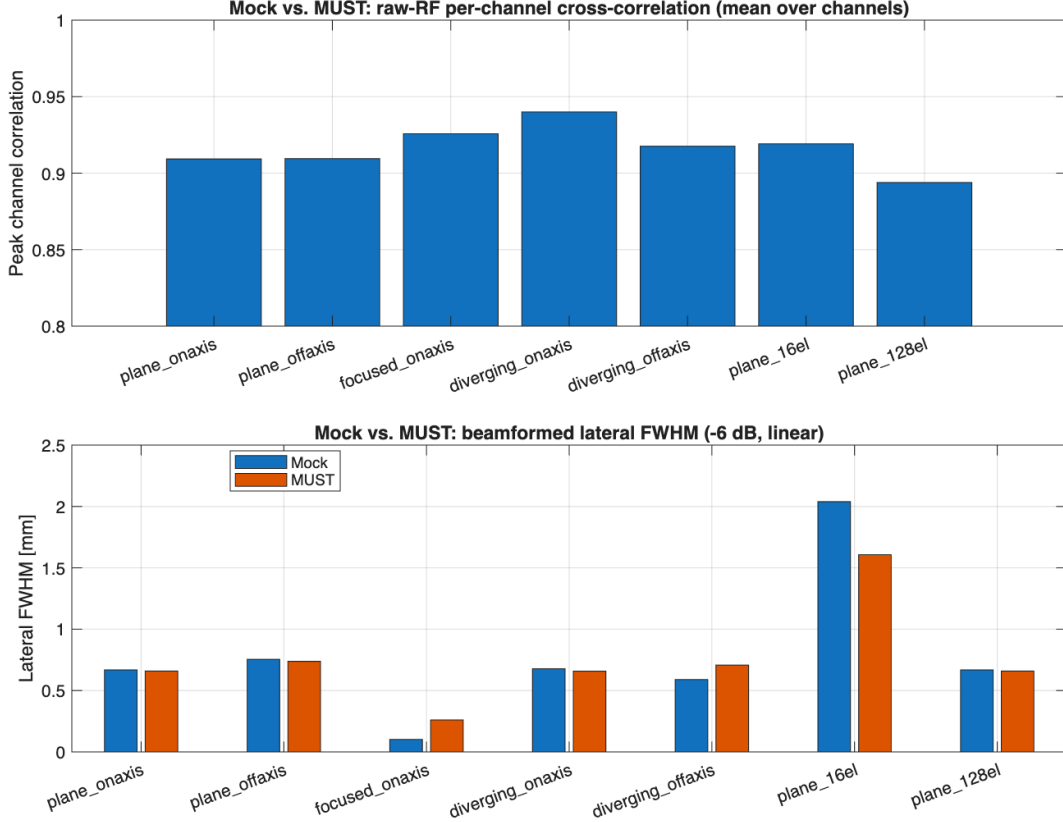


Figure 5: Per-case summary. Top: raw-RF per-channel correlation. Bottom: lateral FWHM, mock vs. MUST. The focused-on-axis case is the clear outlier, discussed in the text.

- MUST itself was only exercised on small, few-scatterer scenes for this validation (Section 5); the reported per-call timing is not representative of large, clinically realistic scenes.

7 Availability

The source code, the validation scripts and results, and the $\text{\LaTeX}$ source of this manuscript are available at <https://github.com/RyoMurakami-MedRobo/ultrasound-edu> under the MIT license. MUST itself is a separate dependency, licensed by its authors under the GNU LGPLv3 and not bundled with this repository; a helper script (`setup_must.m`) downloads it from its official distribution point on request.

8 Conclusion

We presented an open, dependency-light educational tool for exploring and benchmarking delay-and-sum ultrasound beamforming, built so that a learner can change the acquisition, see the transmit wavefront and the exact samples a beamformer sums as explicit geometric objects, and drop in their own algorithm under a single documented interface for quantitative comparison against a reference implementation and against MUST’s built-in beamformer. Because the tool’s no-dependency fallback RF generator is the only path guaranteed to run everywhere, we validated it quantitatively against MUST and found it timing-accurate and geometrically faithful, with one clearly identified

and quantified exception — focused-transmit lateral resolution — that we report as a limitation rather than omit.

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